Unit I – Learning Problems – Perspectives and Issues - Concept Learning - Version Spaces & Candidate Elimination - Inductive Bias - Decision Tree Learning - Representation Algorithm
- Heuristic Space Search

Rubrics (Applicable to all Assignments)

Criterion	**Excellent (5)**	**Good (4)**	**Satisfactory (3)**	**Needs Improvement (2)**	**Poor (1)**
Problem Understanding	Clearly identifies problem and objectives	Mostly clear	Adequate	Limited understanding	Incorrect understanding
Data Collection & Preparation	Appropriate, clean dataset with preprocessing	Minor preprocessing issues	Basic preprocessing	Poor preprocessing	No preprocessing
Analysis using NumPy/Pandas	Correct and efficient analysis	Mostly correct	Basic analysis	Limited analysis	Incorrect analysis
Visualization & Interpretation	Appropriate graphs with meaningful insights	Good visualization	Basic charts	Minimal interpretation	No visualization
Innovation & SDG Mapping	Excellent real-world relevance with SDG linkage	Good linkage	Moderate linkage	Weak linkage	No SDG relevance
Documentation & Conclusion	Professional report with justified conclusions	Good report	Adequate report	Poor report	Incomplete

Total = 30 Marks

Unit I Analytical Assignments with SDGs

A	Assignment Question	Related SDG
1	Explain how Machine Learning can identify poverty-prone regions using socio-economic data.	**SDG 1 – No Poverty**
2	Designa Machine Learning model to identify eligible beneficiaries for government welfare schemes.	**SDG 1 – No Poverty**
3	Explain how Machine Learning can predict crop yield using historical agricultural data.	**SDG 2 – Zero Hunger**
4	Develop a Machine Learning solution for crop disease detection using leaf images.	**SDG 2 – Zero Hunger**
5	Explain how Machine Learning can reduce food wastage in supermarkets.	**SDG 2 – Zero Hunger**
6	Explain the role of Machine Learning in disease prediction and early diagnosis.	**SDG 3 – Good Health and Well-being**
7	Design a Machine Learning model for diabetes prediction.	**SDG 3 – Good Health and Well-being**
8	Explain how Artificial Intelligence assists doctors in medical image analysis.	**SDG 3 – Good Health and Well-being**
9	Design a Machine Learning model to predict student academic performance.	**SDG 4 – Quality Education**
10	Explain how recommendation systems improve online learning platforms.	**SDG 4 – Quality Education**
11	Develop an intelligent attendance prediction system using Machine Learning.	**SDG 4 – Quality Education**
12	Explain how Machine Learning helps detect gender bias in recruitment.	**SDG 5 – Gender Equality**
13	Design an AI-based application to promote equal educational opportunities.	**SDG 5 – Gender Equality**
14	Explain how Machine Learning predicts water quality using sensor data.	**SDG 6 – Clean Water and Sanitation**

15	Develop a Machine Learning model to detect water contamination.	**SDG 6 – Clean Water and Sanitation**

A	**Assignment Question**	**Related SDG**
16	Explain electricity demand forecasting using Machine Learning.	**SDG 7 – Affordable and Clean Energy**
17	Design a Machine Learning model for solar energy generation prediction.	**SDG 7 – Affordable and Clean Energy**
18	Explain how Machine Learning predicts employment trends.	**SDG 8 – Decent Work and Economic Growth**
19	Design a Machine Learning-based job recommendation system.	**SDG 8 – Decent Work and Economic Growth**
20	Explain predictive maintenance in industries using Machine Learning.	**SDG 9 – Industry, Innovation and Infrastructure**
21	Design a Machine Learning-based quality inspection system for manufacturing.	**SDG 9 – Industry, Innovation and Infrastructure**
22	Explain how Machine Learning improves financial inclusion.	**SDG 10 – Reduced Inequalities**
23	Design an AI application for differently-abled people using Machine Learning.	**SDG 10 – Reduced Inequalities**
24	Explain traffic congestion prediction using Machine Learning.	**SDG 11 – Sustainable Cities and Communities**
25	Design a smart parking prediction system using Machine Learning.	**SDG 11 – Sustainable Cities and Communities**
26	Explain waste collection optimization using Machine Learning.	**SDG 11 – Sustainable Cities and Communities**

27	Explain demand forecasting using Machine Learning in retail industries.	**SDG 12 – Responsible Consumption and Production**

A	**Assignment Question**	**Related SDG**
28	Design a Machine Learning system to reduce industrial waste.	**SDG 12 – Responsible Consumption and Production**
29	Explain weather forecasting using Machine Learning techniques.	**SDG 13 – Climate Action**
30	Design a Machine Learning model for flood prediction.	**SDG 13 – Climate Action**
31	Explain forest fire detection using Artificial Intelligence and Machine Learning.	**SDG 13 – Climate Action**
32	Explain marine pollution monitoring using Machine Learning.	**SDG 14 – Life Below Water**
33	Design a fish species classification system using Deep Learning.	**SDG 14 – Life Below Water**
34	Explain wildlife monitoring using Machine Learning.	**SDG 15 – Life on Land**
35	Design a Machine Learning model for deforestation detection.	**SDG 15 – Life on Land**
36	Explain fraud detection using Machine Learning.	**SDG 16 – Peace, Justice and Strong Institutions**
37	Design a crime prediction system using Machine Learning.	**SDG 16 – Peace, Justice and Strong Institutions**
38	Explain the importance of data sharing in achieving Sustainable Development Goals using Machine Learning.	**SDG 17 – Partnerships for the Goals**

39	Design a collaborative Artificial Intelligence platform to solve SDG-related problems.	**SDG 17 – Partnerships for the Goals**
40	Select any one Sustainable Development Goal and propose a complete Machine Learning solution including the problem statement, dataset, algorithm, implementation steps, expected outcome, and societal benefits.	**Any SDG**

Expected Deliverables for Each Assignment

- Problem statement and identification of the relevant Sustainable Development Goal (SDG).
- Background study of the selected application domain and its Machine Learning challenges.
- Identification of the learning problem, input features, target output, and suitable data representation.
- Selection and justification of an appropriate Machine Learning approach (Concept Learning, Decision Tree, Version Space, or Heuristic Search).
- Explanation of the proposed algorithm/model with a workflow or decision-making process.
- Analysis of the expected outcomes, advantages, limitations, and practical applications.
- Recommendations for solving the selected problem aligned with the chosen SDG.
- Conclusion and references.

These assignments are designed to achieve the **Unit I learning outcomes** by enabling students to understand the fundamentals of Machine Learning, including learning problems, concept learning, version spaces, inductive bias, decision tree learning, representation techniques, and heuristic search. The assignments integrate the **United Nations Sustainable Development Goals (SDGs)** through real-world problem-solving, encouraging students to apply Machine Learning concepts to societal and industrial challenges.

6. Problem Statement

Early diagnosis of diseases is essential for effective treatment and reducing mortality rates. Traditional diagnosis depends on medical experts and laboratory tests, which may be time-consuming and expensive. Machine Learning (ML) can analyze patient health records and predict diseases at an early stage, helping doctors make faster and more accurate decisions.

Relevant SDG: SDG 3 – Good Health and Well-being

2. Background Study

Disease prediction is one of the major applications of Machine Learning in healthcare. Hospitals generate huge amounts of patient data, including age, symptoms, blood pressure, glucose level, cholesterol level, medical history, and lifestyle factors.

Machine Learning algorithms learn patterns from historical patient data and predict whether a person is likely to develop a disease such as:

- Diabetes
- Heart Disease
- Cancer
- Kidney Disease
- Liver Disease

Early prediction improves treatment success and reduces healthcare costs.

3. Learning Problem

Learning Type

Supervised Learning (Classification)

Input Features

- Age
- Gender
- Blood Pressure
- Blood Sugar
- Cholesterol
- BMI
- Heart Rate
- Smoking Status
- Family History
- Symptoms

Target Output

- Disease Present (Yes)
- Disease Absent (No)

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4. Data Representation

The dataset is represented as a table.

Age	BP	Sugar	Cholesterol	BMI	Disease
45	130	160	220	29	Yes
32	120	95	180	24	No
55	145	180	240	31	Yes
28	110	90	170	22	No

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5. Machine Learning Approach

The suitable ML approaches are:

- Decision Tree
- Naïve Bayes
- Logistic Regression
- Random Forest

Chosen Algorithm: Decision Tree

Why Decision Tree?

- Easy to understand
- Fast prediction
- High accuracy
- Handles both numerical and categorical data
- Suitable for medical diagnosis

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6. Workflow

```
Patient Data
       │
       ▼
Data Preprocessing
       │
       ▼
Feature Selection
       │
       ▼
Decision Tree Training
       │
       ▼
Disease Prediction
       │
       ▼
Doctor Decision Support
```

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7. Decision Tree Example

```
              Blood Sugar >140?
                /          \
             Yes            No
             /               \
      Cholesterol>200?     Healthy
          /      \
       Yes        No
       /           \
Disease       Healthy
```

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8. NumPy and Pandas Analysis

Machine Learning libraries used:

- NumPy
- Pandas
- Scikit-learn

Example operations:

- Loading dataset
- Removing missing values
- Splitting training and testing data
- Model training
- Prediction

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9. Visualization

Graphs used:

- Age Distribution
- Disease Count
- Correlation Heatmap
- Confusion Matrix
- Accuracy Graph

These graphs help doctors understand disease trends.

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10. Expected Outcome

The system predicts whether a patient is likely to have a disease.

Example:

```
Input

Age = 52
Sugar = 180
BP = 145
BMI = 31

Prediction

Disease = YES
Confidence = 96%
```

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11. Advantages

- Early disease detection
- High prediction accuracy
- Faster diagnosis
- Reduces healthcare cost
- Assists doctors in treatment planning

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12. Limitations

- Requires high-quality data
- May produce incorrect predictions with insufficient data
- Needs regular model updates
- Cannot completely replace doctors

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13. Practical Applications

- Hospitals
- Diagnostic Centers
- Health Insurance Companies
- Telemedicine Platforms
- Smart Healthcare Systems

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14. Recommendations

- Collect larger healthcare datasets.
- Use deep learning for medical image analysis.
- Integrate wearable device data.
- Maintain patient privacy and data security.
- Regularly retrain models with new data.

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15. Conclusion

Machine Learning plays a significant role in disease prediction and early diagnosis by analyzing patient data efficiently. Decision Tree models help identify diseases accurately and support healthcare professionals in making timely decisions. This contributes to **SDG 3 – Good Health and Well-being** by improving healthcare quality and saving lives.

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16. References

1. Tom M. Mitchell, *Machine Learning*, McGraw-Hill.
2. Ian Goodfellow, *Deep Learning*.
3. UCI Machine Learning Repository – https://archive.ics.uci.edu/
4. Scikit-learn Documentation – https://scikit-learn.org/
5. World Health Organization (WHO) – https://www.who.int/

This answer covers all the expected deliverables in your assignment rubric, including the problem statement, SDG mapping, learning problem, data representation, ML approach, workflow, expected outcomes, advantages, limitations, recommendations, conclusion, and references.